

Notes: Brown, Ivkovic, Smith, and Weisbenner (JF, 2008)

Neighbors Matter: Causal Community Effects and Stock Market Participation

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1 Big picture

- **Question.** Does living in a community with more stock market participants *causally* make an individual more likely to participate in the stock market?
- **Setting.** The paper studies U.S. taxable stock ownership using a 10-year panel of taxpayers from 1987 to 1996. The unit of analysis is the taxpayer-year, and the relevant peer group is the local *community*, defined as the Metropolitan Statistical Area (MSA).
- **Core challenge.** A simple correlation between own stockholding and average community stockholding is not informative. People sort into communities, and communities can be hit by common information or economic shocks. Both forces can generate a spurious positive relationship even if there are no peer effects.
- **Main empirical strategy.** The paper restricts attention to *native nonmovers* and instruments community ownership with the lagged average stock ownership of the birth states of *nonnative neighbors*. Combined with household fixed effects, rich time-varying individual and community controls, and state-year fixed effects, this gives a credible source of exogenous variation in local stock market participation.
- **Main result.** A 10-percentage-point increase in average community ownership raises an individual's probability of owning stock by about **4.2 percentage points**.
- **Mechanism evidence.** The effect is stronger in more **sociable** communities, measured using survey evidence on whether neighbors often ask each other for advice about products and brands. This is consistent with a word-of-mouth channel.
- **Economic significance.** The estimated community effect is comparable in magnitude to the effect of one's **parents** switching from nonparticipation to participation in the stock market.
- **Economic takeaway.** Financial participation has a social component. Policies that increase stockholding or financial sophistication may have multiplier effects through local interaction, not just direct effects on the targeted households.

2 Data and measurement (key objects)

2.1 Observed taxpayer-level data

- The paper uses a large **IRS panel of tax returns** covering **1987–1996**. The panel starts from the 1987 IRS cross section of about 88,000 returns and follows those taxpayers through time.
- The final overall sample contains **753,521 taxpayer-year observations** on **85,888 distinct taxpayers**.
- The tax data include detailed information from Form 1040 and associated schedules, plus Social Security information such as age, gender, and the state in which a taxpayer applied for a Social Security number.
- The latter variable is crucial because it gives the taxpayer's **birth state** in the paper's terminology: the state in which the individual resided when the Social Security number was issued.

- The tax data are matched to external sources:
 - Census data for community demographics and time trends,
 - Compustat and CRSP for characteristics of publicly traded firms headquartered in the community,
 - DDB Life Style survey data for community sociability.

2.2 Stock market participation measure

- The paper measures equity ownership using whether the taxpayer **reports dividends** on the federal tax return in that year.
- This captures stock and taxable equity mutual fund ownership in **taxable accounts**. It does *not* capture holdings only inside retirement accounts.
- The dividend proxy is imperfect:
 - it can include payouts from non-equity mutual funds,
 - it may miss non-dividend-paying stocks, especially growth stocks.
- The authors validate the proxy using the Survey of Consumer Finances (SCF): over the overlapping period, the trend in reported dividends tracks the trend in actual stockholding fairly well, and the correlation between actual and proxied ownership is about **0.62**.

2.3 Markets, community, and sample splits

- A **community** is an **MSA**. This is large enough to capture neighbors and coworkers, but still local enough to represent a meaningful peer group.
- There are **337 MSAs**. Individuals outside a formal MSA are assigned to the nearest MSA.
- The paper partitions observations into three groups:
 - **Natives**: taxpayers whose current state of residence is their birth state and who stay in the same MSA over the sample.
 - **Nonnatives**: taxpayers whose current MSA does not overlap with their birth state.
 - **Others**: everyone else.
- The core analysis sample is the native nonmover sample with **398,585 taxpayer-year observations**. Because the instrument is lagged one year, the IV regressions use **353,281 observations**.
- When constructing community ownership, the paper uses the full tax sample of 753,521 observations. When constructing the instrument, it uses the **220,783 nonnative observations**.

2.4 Key descriptive facts

- **Own stock ownership rate**: the mean ownership rate in the pooled sample is **25.6%**.
- **Community ownership**: the median taxpayer lives in a community where **26%** of residents own stock; the 10th percentile is about **16.6%** and the 90th percentile is about **34.0%**.

- **Birth-state ownership:** the ownership rate in an individual's birth state has mean **25.6%** and somewhat less dispersion than community ownership.
- **Instrument distribution:** the average ownership of nonnative neighbors' birth states has mean about **25.5%**, median **25.1%**, and 90th percentile around **29.9%**.
- **Income and age:** mean noncapital income is about **\$43.7 thousand** and median noncapital income is about **\$31.4 thousand**; mean age is **46.3**.

2.5 Unobservables and timing

- The main econometric threat is unobserved heterogeneity in the form of sorting into communities on characteristics like risk tolerance, ideology, social capital, or nonlinear functions of observed traits.
- The paper also worries about **time-varying community shocks**: new brokers, local financial media, local booms, or other common changes in the information environment.
- Timing matters because the instrument is lagged one year. This is meant to isolate community-level changes that affect own stock ownership with a delay through neighbor interactions rather than through a simultaneous response to national news.

3 Models and empirical specifications (all equations + interpretation)

3.1 Why simple OLS fails

- Let $y_{i,m,t} \in \{0, 1\}$ denote whether individual i in community m owns stock at time t .
- Define average ownership of the *other* members of the community by

$$\bar{y}_{i,m,t} = \frac{1}{I_{m,t} - 1} \sum_{j \neq i} y_{j,m,t}.$$

- If individuals sort into communities based on an omitted trait U that also affects stockholding, then in community m each person effectively has participation probability p_m .
- In that case, with many observations in the MSA, $\bar{y}_{i,m,t} \approx p_m$, and a regression of $y_{i,m,t}$ on $\bar{y}_{i,m,t}$ is approximately a regression of $y_{i,m,t}$ on p_m .
- Since

$$\text{Cov}(y_{i,m,t}, p_m) = \text{Var}(p_m),$$

the coefficient can be strongly positive even if there is *no* causal peer effect.

- **Interpretation.** The problem is not just Manski's reflection issue. It is also sorting on unobservables and common shocks. A large OLS coefficient can be entirely noncausal.

3.2 The core second-stage regression

- The empirical specification is a linear probability model, with the dependent variable rescaled to $\{0, 100\}$ so coefficients are directly interpretable as percentage-point effects.
- A stylized version of the main equation is

$$y_{i,m,t} = \beta \widehat{\text{Own}}_{m,t} + X'_{i,t}\Gamma + C'_{m,t}\Lambda + \alpha_i + \delta_{s,t} + u_{i,m,t},$$

where:

- $\widehat{\text{Own}}_{m,t}$ is instrumented average community ownership,
- $X_{i,t}$ are time-varying household controls,
- $C_{m,t}$ are time-varying community controls,
- α_i are household fixed effects,
- $\delta_{s,t}$ are state-year fixed effects.
- Because the sample is restricted to **nonmovers**, household fixed effects also absorb time-invariant community fixed effects.
- **Interpretation.** Identification comes from within-household changes in own stockholding tied to within-community changes in community stockholding, after stripping out a broad set of household, community, and state-year confounds.

3.3 Instrument construction

- The instrument is the **1-year-lagged average stock ownership of the birth states of nonnative community members**. A stylized version is

$$Z_{m,t-1} = \frac{1}{N_{m,t}^{NN}} \sum_{j \in NN(m,t)} \text{Own}_{b(j),t-1},$$

where:

- $NN(m,t)$ is the set of nonnative residents in community m at time t ,
- $b(j)$ is person j 's birth state,
- $\text{Own}_{b(j),t-1}$ is stock ownership in that birth state in the previous year.
- Intuition through the paper's St. Louis example:
 - if a St. Louis native has a neighbor from Massachusetts, Massachusetts stockholding may predict the neighbor's propensity to own stock;
 - but for a St. Louis native, Massachusetts stockholding should not directly affect own stockholding except through neighbor interaction.
- The first-stage equation is conceptually

$$\text{Own}_{m,t} = \pi Z_{m,t-1} + X'_{i,t}\Gamma + C'_{m,t}\Lambda + \alpha_i + \delta_{s,t} + \eta_{i,m,t}.$$

- **Interpretation.** The instrument shifts community ownership through the financial background of nonnative neighbors, while the native-sample restriction is designed to keep it orthogonal to own background.

3.4 First-stage evidence and decomposition

- In the simple first stage with no other controls, the coefficient on the lagged instrument is **49.5** with an F-statistic of **20.4**.
- With the full set of controls, household fixed effects, and state-year fixed effects, the coefficient is still **24.6** and the first-stage F-statistic is **10.6**.
- The instrument varies through two channels:
 1. **Mover channel:** community composition changes, holding birth-state ownership fixed.
 2. **Stayer channel:** the birth-state ownership of the same nonnative neighbors changes over time.
- When entered separately, the mover component has coefficient **24.2** and the stayer component coefficient **15.0**; both matter, and the paper cannot reject equality of their effects.
- **Interpretation.** The first stage is not driven by just one mechanical source of variation. Both community composition and changes in the birth-state ownership environment of the same nonnative neighbors contribute to relevance.

3.5 Sociability interaction

- To test the word-of-mouth mechanism more directly, the paper estimates

$$y_{i,m,t} = \beta \widehat{\text{Own}}_{m,t} + \theta S_{m,t} + \gamma \widehat{\text{Own}}_{m,t} S_{m,t} + X'_{i,t} \Gamma + C'_{m,t} \Lambda + \alpha_i + \delta_{s,t} + u_{i,m,t},$$

where $S_{m,t}$ is a normalized measure of community sociability.

- The sociability measure is based on agreement with the statement:

“My friends and neighbors often come to me for advice about products and brands.”

- The key interaction coefficient is **47.3** (significant at the 10% level), implying that community effects are stronger where neighbor advice-seeking is more prevalent.
- **Interpretation.** If the main coefficient came only from omitted common shocks, it should not systematically be stronger in more sociable communities. The interaction result therefore supports a social-interaction channel.

3.6 Economic significance benchmark

- The baseline IV estimate is

$$\beta \approx 41.8,$$

where the dependent variable is scaled to $\{0, 100\}$.

- Therefore, a 10-percentage-point increase in community ownership implies an increase in own participation of about

$$0.10 \times 41.8 \approx 4.2$$

percentage points.

- The paper compares this with a parent-child benchmark using a linked dependent sample. If one's parents switch from not owning stocks to owning stocks, own participation rises by about **3.6 percentage points**.
- **Interpretation.** The estimated community effect is not just statistically significant; it is economically large and comparable to a major family influence.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- A simple regression of own stock ownership on average community ownership is contaminated by:
 - sorting into communities,
 - omitted individual traits,
 - omitted community traits,
 - common time-varying shocks to information or economic conditions.
- Even panel data with many controls are not enough by themselves, because there can still be time-varying omitted shocks at the community level.

4.2 What identifies the causal effect

- The instrument is relevant because birth-region financial culture or social capital has persistent effects on the stockholding behavior of migrants.
- The instrument is plausibly excluded because, for native nonmovers, stockholding in the birth states of nonnative neighbors should matter for own stockholding only through neighbors' own behavior.
- The lagged instrument helps rule out contemporaneous common responses to national information events.

4.3 Why the specification is unusually demanding

- **Household fixed effects** remove time-invariant observed and unobserved individual traits.
- Since the sample is restricted to nonmovers, these same household fixed effects also remove time-invariant community characteristics.
- **Time-varying household controls** remove changes in income, age, filing status, homeownership proxies, self-employment, supplemental income, and retirement saving behavior.
- **Time-varying community controls** remove shifts in local demographics, education, occupation, industry mix, median home values, population, and the characteristics of locally headquartered firms.

- **State-year fixed effects** absorb any shocks that are common at the state-year level, including state-specific booms, recessions, tax changes, or changes in local norms that are state-wide rather than MSA-specific.

4.4 Remaining identifying assumption and limits

- The key remaining assumption is that lagged ownership in the birth states of nonnative neighbors affects natives' stockholding only via local social interaction and not through omitted MSA-specific time-varying shocks that are correlated with migrant origins.
- The authors acknowledge a possible alternative story: communities with changing industry composition might attract migrants from similar communities elsewhere. They argue this is mitigated by the large set of controls for local economic conditions and by state-year effects.
- The equity ownership proxy is limited to taxable accounts and dividend-paying assets. This creates measurement error, but the SCF validation suggests it is informative and the time pattern is reasonable.

4.5 Relation to the literature

- Relative to **Hong, Kubik, and Stein (2004)**, the paper provides a cleaner causal design rather than relying on a direct correlation between sociability proxies and participation.
- Relative to **Guiso, Sapienza, and Zingales (2004)**, the paper does not just link social capital to finance; it identifies a community-level peer effect on stockholding.
- Relative to **Feng and Seasholes (2004)**, who emphasize common reactions to public information, this paper argues that the data are more consistent with causal word-of-mouth spillovers.

5 Tables and key empirical evidence

5.1 Table I: summary statistics

Read:

- Table I establishes that taxable stock market participation is common but far from universal: only **25.6%** of the pooled taxpayer sample owns stock by the dividend measure.
- There is substantial cross-community dispersion in stockholding: taxpayers in the 90th-percentile community live around a **34%** ownership rate, versus about **16.6%** in the 10th-percentile community.
- This dispersion is exactly what makes the question interesting, but by itself it says nothing about causality.
- The table also clarifies the scale of the sample and the richness of household covariates available from tax records.

Table II
First-Stage Regression of Community Equity Ownership,
1988 to 1996 Pooled

The dependent variable, individuals' average community stock ownership, is measured in percentage points, whereas all explanatory variables are expressed as raw ratios (i.e., not in percentage points). The sample is restricted to native individuals (those whose current state of residence is their birth state and who have lived in the same MSA for the entire sample period). Birth state refers to the state in which individuals lived when they applied for their Social Security numbers (generally the state in which they were born). Nonnatives are those whose Social Security numbers were issued in states that do not overlap with their current community. Community is defined as MSA. The key independent variable is the average equity ownership of nonnative community members' birth states. It is lagged by 1 year to reduce the potential for picking up a spurious correlation that might arise if, for example, there were a simultaneous correlated response across states to the release of new information. Other controls (utilized in Column (2)) include time-varying household-level controls (specifically, (i) an indicator variable for each percentile of the noncapital income distribution as measured each year, for a total of 99 variables, with the 50th percentile as the omitted category, (ii) an indicator variable for each possible age through age 106 (the oldest in our sample), with age 40 as the omitted category, (iii) indicator variables for gender interacted with filing status—single male, single female, male head of household, female head of household, and (iv) numerous other items from the tax return, including indicators for whether the household claims a mortgage interest deduction, self-employment, supplemental income on Schedule E, contributions made to tax-qualified defined contribution plans, as well as the number of dependents), household fixed effects (and, because the sample is restricted to non-movers, the household fixed effects also control for community fixed effects), state-year fixed effects, and a number of time-varying community-level controls. The latter include community-level averages of all of the individual-level controls for income, age, filing status, and so on (for age and income, we include a quadratic specification of the community averages). In addition, we use 1980, 1990, and 2000 Census data to compute community-specific time trends for a host of variables, including the distribution of completed education in the community (i.e., fraction of community that completed grades 0–8, 9–11, high school, some college, and college graduates); the distribution of community residents by race and ethnicity (i.e., the fraction of Hispanics, Blacks, Indians, Asians, and other groups in the MSA); and the distribution across 13 occupation classifications, 11 industry classifications, and six sector classifications (e.g., state and local government, private sector). We also include trends in community median home value to capture community-specific economic conditions, as well as nonlinear controls for the MSA population each year. Finally, we also include a large set of covariates characterizing locally headquartered, publicly traded firms (an indicator whether there are publicly traded firms headquartered within the community, the fraction of total U.S. market value represented by these local firms, the value-weighted rate of return of local firms, the fraction of local firms that were in the top 10% of nationwide performers that year, the fraction of local firms that pay dividends, and the fraction of the population employed by these local firms). Standard errors, shown in parentheses, are estimated by clustering on the MSA level. Thus, these standard errors allow for correlation across individuals within each community for a given year, as well as correlation across individuals in the same MSA at different points in time. Moreover, because we limit the sample to natives, our standard error calculation also allows for correlation in observations of the same individual over time. The standard errors also allow for heteroskedasticity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1) No Other Controls	(2) + Other Controls (Including Household Fixed Effects and State-Year Fixed Effects)
Average equity ownership of nonnative community members' birth states (lagged 1 year)	49.5*** (11.0)	24.6*** (7.6)
F-statistic of instrument in first stage	20.4	10.6
Adjusted R ²	0.051	0.928
Number of observations	353,281	353,281

Table I
Summary Statistics for the 1987–1996 Pooled Sample

Statistics are based on a panel of tax returns from 1987 to 1996 and are weighted by population weights. Community is defined as Metropolitan Statistical Area (MSA). Birth state refers to the state in which individuals lived when they applied for their Social Security numbers (generally the state in which they were born). Nonnative community members are those whose Social Security numbers were issued in states that do not overlap with their current community. Panel A focuses on equity ownership (as proxied by reporting dividends on tax returns for the year). Panel B focuses on the remaining individual characteristics our specifications employ in subsequent analyses.

Panel A: Equity Ownership Summary Statistics								
	Mean (Std. Dev.)	1st	10 th	25 th	50 th	75 th	90 th	99 th
Does individual own equity?	25.6 (43.6)							
Average equity ownership of community	25.6 (6.7)	9.7	16.6	21.2	26.0	29.9	34.0	41.8
Ownership of individual's birth state	25.6 (5.4)	13.8	18.1	22.1	26.2	29.4	32.7	37.7
Average equity ownership of nonnative community members' birth states	25.5 (3.2)	17.8	21.9	23.2	25.1	27.8	29.9	32.7

Panel B: Individual Characteristics								
Income (noncapital), 1996\$ (in thousands)	43.7 (195.2)	–4.5	1.7	12.8	31.4	58.1	89.5	243.9
Age	46.3 (16.2)	20	27	33	43	58	70	86
Single filing status—male* (in%)	18.6 (38.9)							
Single filing status—female* (in%)	16.0 (36.7)							
Head of HH filing status—male* (in%)	4.0 (19.7)							
Head of HH filing status—female* (in%)	7.0 (25.6)							
Claim mortgage interest deduction (in%)	31.4 (46.4)							
Number of dependents	0.8 (1.1)	0	0	0	0	1	2	4
Self-employed (in%)	6.2 (24.2)							
File Schedule E—supplemental income (in%)	15.9 (36.6)							
Pay Alternative Minimum Tax (in%)	0.3 (5.1)							
Contribute to DC plan (in%)	34.7 (47.6)							

*Married couples filing jointly is the omitted category, accounting for 52.4% of all observations. Married couples filing separately are excluded from the sample.

5.2 Table II: first-stage regression

Read:

- Table II shows that the lagged birth-state-ownership instrument is relevant.
- Without controls, the first-stage coefficient is **49.5**; with the full battery of controls and fixed effects, it remains **24.6**.
- The first-stage F-statistics of **20.4** and **10.6** suggest that the instrument is strong enough to be useful in the IV design.
- The persistence of first-stage relevance after the demanding control set is important: the instrument is not just picking up obvious differences across communities.

5.3 Table III: the core IV result

Read:

- Column (1) is the headline estimate. The coefficient on community ownership is **41.8**, implying that a 10-percentage-point increase in community ownership raises own participation by about **4.2 percentage points**.
- Column (2) interacts instrumented community ownership with sociability. The interaction coefficient is **47.3**, showing that the effect is stronger where neighbors are more likely to exchange

Table III
Instrumental Variables Regression, 1988 to 1996 Pooled

The dependent variable, individuals' stock ownership, is measured in percentage points, whereas all explanatory variables are expressed as raw ratios (i.e., not in percentage points). The sample is restricted to native individuals (i.e., those whose current state of residence is their birth state and who have lived in the same MSA for the entire sample period). Equity ownership within the community is instrumented for with the 1-year-lagged average equity ownership of nonnative community members' birth states (for details concerning the first-stage regression, see Section III.B and Table II). Birth state refers to the state in which individuals lived when they applied for a Social Security number (generally the state in which they were born). Nonnative community members are community members who were issued Social Security numbers in states that do not overlap with their current community. Both specifications include household fixed effects (and, because the sample is restricted to nonmovers, the household fixed effects also control for community fixed effects), time-varying household-level controls, time-varying community-level controls, and state-year fixed effects. The time-varying household-level controls and time-varying community-level controls are the same as those included in Table II (see Section IV. A for their description). The Community asks for advice variable, utilized in Column (2), is constructed on a community-year basis using the DDB Life Style survey respondents' perceptions of how often they are asked for advice regarding products and brands (see Section IV. C for further details). This variable is normalized to range from zero to one, with zero being least likely to be asked by neighbors for advice and one being most likely to be asked by neighbors for advice. Community is defined as MSA. Standard errors, shown in parentheses, are estimated by clustering on the MSA level. Thus, these standard errors allow for correlation across individuals within each community for a given year, as well as correlation across individuals in the same MSA at different points in time. Moreover, because we limit the sample to natives, our standard error calculation also allows for correlation in observations of the same individual over time. The standard errors also allow for heteroskedasticity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
Equity ownership within community	41.8*** (15.6)	21.5 (23.2)
Community asks for advice		-13.1 (8.6)
Equity ownership within community × Community asks for advice		47.3* (27.3)
Adjusted R^2	0.781	0.778
Number of observations	353,281	304,107

advice.

- The baseline community coefficient in column (2) drops to **21.5** and is imprecisely estimated because the effect is now allowed to vary with sociability.
- The table therefore provides both the **causal estimate** and the main piece of **mechanism evidence**.

5.4 Economic significance in the text

Read:

- The paper's parent-child benchmark is especially useful for scale. A change in parents' participation status raises children's participation by about **3.6 percentage points**.
- Since the community effect from moving from the 25th to the 75th percentile of community ownership is about **4.2 percentage points**, the implied peer effect is economically large.
- This gives the main policy message teeth: peer-based multipliers in financial participation are plausibly meaningful in real-world magnitudes, not just statistically nonzero.

6 Short summary

- **Method.** The paper uses a taxpayer panel, native-nonmover sample restriction, and an instrument based on the lagged stockholding of the birth states of nonnative neighbors.
- **Identification.** Household fixed effects, community fixed effects, rich time-varying controls, and state-year effects remove most standard confounds; the instrument supplies exogenous variation in local participation.

- **Main result.** A 10-percentage-point increase in community stock market participation raises own participation by about **4.2 percentage points**.
- **Mechanism.** The effect is stronger in more sociable communities, supporting a word-of-mouth explanation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a causal community effect in stock market participation, not merely a correlation.
- Individuals are more likely to enter the stock market when the people around them are more likely to invest in stocks.
- The paper also shows that the most plausible mechanism is social interaction and information transmission rather than a common omitted shock.

7.2 Economic intuition

- Investing in stocks is not purely an isolated optimization problem. It also depends on what people learn from those around them.
- Neighbors can lower the psychological and informational fixed costs of investing by sharing knowledge about accounts, products, and market participation.
- Because of this, participation can exhibit local multiplier effects: once enough people in a community participate, it becomes easier for others to do the same.

7.3 Takeaway

This paper brings a difficult peer-effects question into a credible identification framework. For household finance, the main lesson is that participation in risky asset markets is socially transmitted. For policy, the implication is that financial education or participation-oriented reforms may spread through communities rather than operate only one household at a time.